

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Fourth Semester B.Tech Degree Examination July 2021 (2019 Scheme)

Course Code: EST200**Course Name: DESIGN AND ENGINEERING**

Max. Marks: 100

Duration: 3 Hours

PART A*Answer all questions. Each question carries 3 marks*

Marks

- | | | |
|----|---|-----|
| 1 | Outline the significance of understanding customer requirements in design process | (3) |
| 2 | Describe any three constraints that can occur in design process of a lunch box | (3) |
| 3 | Explain convergent questioning in design thinking | (3) |
| 4 | Explain how conflict in team environment helps in better design of products | (3) |
| 5 | Describe how prototyping helps in design process | (3) |
| 6 | Explain any three advantages of communicating designs in writing. | (3) |
| 7 | Illustrate advantages of reverse engineering in design | (3) |
| 8 | Explain bio mimicry in design with an example | (3) |
| 9 | Describe ethics to be followed in engineering design | (3) |
| 10 | Explain the significance of sustainability in engineering design | (3) |

PART B*Answer any one full question from each module. Each question carries 14 marks***Module 1**

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|----|---|------|
| 11 | Explain the design process through designing a handbag for women of age group of 15 to 25 years. Use hand sketches to support your idea | (14) |
| 12 | Describe the concept of generating design alternatives and choosing a design through designing a coffee mug with the help of sketches | (14) |

Module 2

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|----|---|------|
| 13 | Illustrate the design thinking process through designing a walking stick for elderly people. | (14) |
| 14 | Design a parachute mechanism for safe landing an egg which is dropped from a height of 3 meters using iterative design thinking process with the help of sketches | (14) |

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Module 3

- 15 Design an office chair and communicate your design using sketches with design detailing, material selection, scale drawings and dimensions (14)
- 16 Describe the role of mathematical modelling in design engineering citing an example (14)

Module 4

- 17 Illustrate modular design approach for designing of desktop computers (14)
- 18 Demonstrate the concept of ergonomics through design of a table lamp (14)

Module 5

- 19 Illustrate the changes in design of disposable tea cup in terms of production, use and sustainability (14)
- 20 Describe the how to estimate the cost of a table in design stage? Show how economics will influence the engineering designs. (14)
